

AI4Math — Lecture 3

Research Updates & Mathematical Practice
Inductive and Quotient Types in Lean 4

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01 News **02** Readings **03** Types & proofs

News snapshot: September 17-23, 2026; includes the September 16 Liouville–Goldbach release.

Preprint / technical replay / peer review: distinct evidence levels.

Research updates 01 | New mathematics

Date	Result	AI role	Evidence / limit
09-16 09-17	Liouville-Goldbach All even integers; unconditional	Public reports: Astra	Lean + replay report; incomplete discovery trace
09-17	Weak Pach-Tardos $\text{Ex}(n, P) \leq n^{1+o(1)}$	Astra attribution: full-text check pending	Abstract checked; preprint
09-17	Talagrand operator cotype Counterexample at $q = 2$	GPT-5.6: counterexample	Explicit disclosure; preprint
09-20	Kahn flow conjecture	Human guidance; Astra proof; author rewrite	Explicit disclosure; preprint

Dates: first release / replay, not journal acceptance.

Sources (click) [\[N1\]](#) [\[N1a\]](#) [\[N1b\]](#) [\[N2\]](#) [\[N3\]](#) [\[N4\]](#)

Research updates 02 | Search and verification

Date	Result	AI / computation	Evidence / limit
09-17	FormalFlow A core theorem behind $\text{MIP}^* = \text{RE}$	Supervised agents: planning, proving, review	Reported: 63 days, 126,367 lines; not all of $\text{MIP}^* = \text{RE}$
09-17	Gardam lattice Failure of unique products	Certificate search; LLM assistance	32×28 products; Lean; torsion-freeness external
09-19	bit-PHP / $\text{Res}(\oplus)$ $2^{\Omega(n/\log^2 n)}$ lower bound	Substantial AI assistance	Paper + Lean; review distinct
09-20	Negative dependence Two counterexamples	Eureka discovery; author verification	$\{0, 1, 2\}^4$; 17,532 exact integer inequalities

Public code $\neq$ rebuilt in this revision. Formalizing a claim $\neq$ first discovering it.

Sources (click) [\[N7\]](#) [\[N8\]](#) [\[N6\]](#) [\[N5\]](#)

Research updates 03 | Watchlist

Date	Item	Content	Status
09-18	Primus for Math	Research system; computation / verification	Product release; evaluation pending
09-20	Magnitude continuity	Finite sets; finite-dimensional L_1 subspaces	New manuscript; secondary AI disclosure
09-21	Grad counterexamples	Magnetohydrostatic equilibria; Lean	Formalization; AI discovery role unconfirmed
09-21	Math & AI advisory group	Review, communication, standards	Official + independent group websites
09-21	OpenAI: 100+ problems	Internal-model capability claim	Aggregate claim; not itemwise verified

Sources (click) [\[N11\]](#) [\[N9\]](#) [\[N9a\]](#) [\[N10\]](#) [\[R2\]](#) [\[R3\]](#)

Liouville-Goldbach | What is weakened?

$$\lambda(n) = (-1)^{\Omega(n)}, \quad \Omega(n) = \text{number of prime factors, with multiplicity.}$$

Claim of the public project

$$\forall N > 2, \quad 2 \mid N \implies \exists a, b \geq 1 : \quad a + b = N, \quad \lambda(a) = \lambda(b) = -1.$$

Classical Goldbach implies this

$$N = p + q \implies \lambda(p) = \lambda(q) = -1.$$

Prime $\Rightarrow$ odd number of prime factors

Sources (click) [\[N1\]](#) [\[N1a\]](#)

AI4Math / Lecture 3

Evidence boundaries

Public proof + Lean artifact

Independent technical replay report

Not a solution of classical Goldbach

Reading the news | Four questions

Mathematics

What was proved?

New theorem / formalized old result / weaker claim

Under which assumptions?

Full range / large parameters / conditional result

Attribution and verification

What did the AI contribute?

Idea / object / proof completion / code

Who checked what?

Author / exact program / kernel / independent reviewer

Certificate $\nrightarrow$ **Faithful specification** $\nrightarrow$ **Clear attribution**

Discussion framework, not a uniform quality ranking of the listed results.

Reading 01 | Gowers

Why I didn't sign the Fields medallists' letter

Timothy Gowers / September 17, 2026

Original excerpt

“there is a spectrum of attitudes in mathematics”

Open the original article

Zhihu reference

Long-form essay; full text on the source website.

Next slide: a paraphrased argument map, not the original text.

Sources (click) [R1] [Gowers's Weblog](#)

Gowers | Different goals, different responses

1. **Shared concern** Crisis acknowledged; no opposing camp claimed
2. **Multiple mathematical temperaments** Problem-solving and understanding
3. **More output** More undigested results, but possibly more absorbed knowledge
4. **Institutional risk** Motivation, training, support for a new generation
5. **Adaptation** Public models: discovery no longer confined to companies

Questions for discussion

Value judgement or capability forecast? Which assumptions drive the conclusion?

Remark | Disagreement and the “silent majority”

More than two camps

Goals: results / understanding / training

Pace: immediate release / careful exposition

Responsibility: credit / review / data boundaries

Support for AI capability need not imply support for every deployment practice.

The “majority” is unmeasured

No signature $\neq$ support for a company

Silence $\neq$ agreement or opposition

Comments $\neq$ representative survey

Unknown size, unknown views: no inference from absence of speech.

Seminar prompts / supplementary analysis

Early-career researchers? Differences across fields? Missing voices?

Reading 02 | OpenAI announcement

Advisory Group on Mathematics and Artificial Intelligence

OpenAI / September 21, 2026

Original excerpt

“The group will operate independently from OpenAI.”

Open the original article

[Independent group website](#)

Short original reading: open during the seminar.

Focus: review, communication, independence, and limits of the remit.

Sources (click) [\[R2\]](#) [\[R3\]](#)

Remark | Independent advice, not decision power

Mandate

Assess significance; advise on review

Coordinate communication and release

Discuss professional standards
Independent members; unpaid by OpenAI

Unsolicited advice; public recommendations

Limits

No company decision-making power

Internal research pace: outside remit

Not blanket validation of results

Advice $\neq$ veto

Decisions and responsibility: the company

Discussion: representation, transparency, uptake, feedback?

Sources (click) [\[R2\]](#) [\[R3\]](#)

From research news to Lean | Two mechanisms

Inductive types

Constructing objects

Defining recursive functions

Proving by structure

build $\longrightarrow$ recurse / prove

Quotient types

Identifying equivalent objects

Representative-independent maps

Proving statements about classes

relation $\longrightarrow$ quotient / lift

Live file: Part A. The supplied Sets & Functions notes remain in Part B of the Lean file only.

Sources (click) [\[L1\]](#) [\[L2\]](#)

Inductive types | Constructors determine shape

```
inductive Tree ( $\alpha$  : Type u) : Type u where  
  | leaf :  $\alpha \rightarrow$  Tree  $\alpha$   
  | node : Tree  $\alpha \rightarrow$  Tree  $\alpha \rightarrow$  Tree  $\alpha$ 
```

Two cases

Leaf: one α -label

Node: two smaller trees

Generated by Lean

Constructors: `Tree.leaf` /

`Tree.node`

Eliminator: `Tree.rec`

Finite binary trees: every value has a constructor-based origin.

Sources (click) [\[L1\]](#)

Recursion | Mirroring a tree

```
def mirror {α : Type u} : Tree α → Tree α
| .leaf a => .leaf a
| .node l r => .node (mirror r) (mirror l)
```

$$\text{mirror}(\text{node}(l, r)) = \text{node}(\text{mirror}(r), \text{mirror}(l)).$$

Why is the recursive definition accepted?

Immediate subtrees $\longrightarrow$ smaller structure $\longrightarrow$ termination

Defining equations reduce; concrete constructor equations often close with `rfl`.

Sources (click) [\[L1\]](#)

Induction | Mirroring twice

```
theorem mirror_mirror {α : Type u} (t : Tree α) :  
  mirror (mirror t) = t := by  
  induction t with  
  | leaf a => rfl  
  | node l r ihL ihR =>  
    simp only [mirror, ihL, ihR]
```

leaf Reduction + rfl

node Two subtrees + two
 hypotheses

cases: split cases **induction: cases + induction hypotheses**

Sources (click) [\[L1\]](#)

Indexed families | Length in the type

```
inductive Vec ( $\alpha$  : Type u) : Nat  $\rightarrow$  Type u where
  | nil : Vec  $\alpha$  0
  | cons {n : Nat} :  $\alpha \rightarrow$  Vec  $\alpha$  n  $\rightarrow$  Vec  $\alpha$  (n + 1)

def Vec.head { $\alpha$  : Type u} {n : Nat} : Vec  $\alpha$  (n + 1)  $\rightarrow$   $\alpha$ 
  | .cons a _ => a
```

Parameter α Element type

Index n Length

Vec α (n+1) excludes the empty vector

head: no nil branch

The input type supplies the constraint; no runtime “nonempty list” check.

Sources (click) [\[L1\]](#)

Two boundaries | Not arbitrary self-reference

Strict positivity

Recursive occurrences are constrained

$$\text{mk} : (\text{Bad} \rightarrow \text{Nat}) \rightarrow \text{Bad}$$

Rejected definition

Negative occurrence: Bad to the left of an arrow inside a constructor argument

Inductive propositions

EvenN : Nat $\rightarrow$ Prop

$$\frac{}{\text{EvenN}(0)} \quad \frac{\text{EvenN}(n)}{\text{EvenN}(n+2)}$$

Constructors build **proof evidence** here

The rejected declaration stays commented out in the demonstration file.

Sources (click) [\[L1\]](#)

Quotients | Identify equivalent objects

$$a \sim b \iff a \bmod 2 = b \bmod 2$$

Natural numbers

0, 1, 2, 3, 4, 5, ...

$2 \neq 4$ Different natural numbers

Quotient

[0]: even class [1]: odd class
[2] = [4] The same equivalence class

The map $q : n \mapsto [n]$ forgets some representative information.

Sources (click) [L2]

Setoid | Package an equivalence relation

```
def modTwo : Setoid Nat where
  r a b := a % 2 = b % 2
  iseqv := {fun _ => rfl,
    fun h => h.symm,
    fun h1 h2 => h1.trans h2}

abbrev Parity := Quotient modTwo

def cls (n : Nat) : Parity := Quotient.mk modTwo n
```

iseqv: reflexive, symmetric, transitive. Parity: the type of equivalence classes.

Sources (click) [\[L2\]](#)

Equality in the quotient | sound / exact

```
theorem two_eq_four : cls 2 = cls 4 :=  
  Quotient.sound (show 2 % 2 = 4 % 2 from by decide)  
  
example (a b : Nat) (h : cls a = cls b) : a % 2 = b % 2 :=  
  Quotient.exact h
```

sound: $a \sim b \Rightarrow [a] = [b]$ exact: $[a] = [b] \Rightarrow a \sim b$

Two notions of equality

rfl: definitional reduction sound: equality from the relation

The proof of $[2] = [4]$ does not reduce the representative 2 to 4.

Sources (click) [\[L2\]](#)

Quotient.lift | The well-definedness obligation

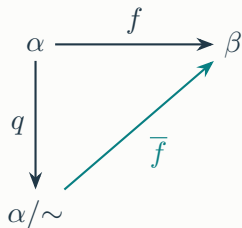
Required

$$\forall a, b, \quad a \sim b \implies f(a) = f(b).$$

Obtained

$$\overline{f}([a]) = f(a).$$

Output independent of the representative.



Sources (click) [\[L2\]](#)

A function that descends | Remainder

```
def residue : Parity → Nat :=  
  Quotient.lift (fun n => n % 2)  
    (by intro a b h; exact h)  
  
example (n : Nat) : residue (cls n) = n % 2 := rfl  
example : residue (cls 7) = 1 := rfl
```

Well-definedness in this example

$a \sim b$ means $a \bmod 2 = b \bmod 2 \implies$ use the hypothesis

Computable remainder 0/1; not recovery of the original arbitrary representative.

Sources (click) [\[L2\]](#)

Quotient induction | Prove on representatives

```
theorem residue_lt_two (q : Parity) : residue q < 2 := by
  refine Quotient.inductionOn q ?_
  intro n
  exact Nat.mod_lt n (by decide)
```

$$\underbrace{\forall q : \mathbb{N}/\sim, P(q)}_{\text{proposition about classes}} \longleftarrow \underbrace{\forall n : \mathbb{N}, P([n])}_{\text{proof on representatives}}$$

An elimination principle, not an arbitrary hidden representative choice.

Sources (click) [\[L2\]](#)

A failed lift | Recovering every input

```
theorem no_preserving_representative :
  ¬ ∃ f : Parity → Nat, ∀ n, f (cls n) = n := by
  rintro ⟨f, hf⟩
  have h := congrArg f two_eq_four
  rw [hf 2, hf 4] at h
  omega
```

$$[2] = [4] \implies f([2]) = f([4]) \implies 2 = 4 \quad \perp$$

Preserving every representative is impossible; choosing a canonical one is not.

Sources (click) [\[L2\]](#)

Quot versus Quotient | One final distinction

	Quot r	Quotient s
Input	Any binary relation r	Equivalence relation: Setoid
Relation implies equality	Quot.sound	Quotient.sound
Equality implies relation	Not generally; equivalence closure	Quotient.exact
Define quotient functions	Quot.lift	Quotient.lift

Quotient equality: reflexive, symmetric, transitive.

Kernel-supported quotients: not an ordinary inductive declaration with arbitrary new equations.

Sources (click) [\[L2\]](#) [\[L3\]](#)

Live demo | Three entry points

1. `namespace InductiveDemo`

Constructors, recursion, changing goals in `mirror_mirror`

2. `namespace QuotientDemo`

Compare `sound`, `lift`, and `inductionOn`

3. `AI4Math_Demo.lean`

Latest uploaded code; condensed source-prose comments in Part B

Before presenting

Lean 4.19.0 + mathlib v4.19.0 / `bash verify.sh`

Comment-only revision; executable source unchanged.

Lean compiler unavailable here: local compilation still required.

Source index 01 | Papers and projects

N1	Liouville–Goldbach: primary repository
N1a	Independent replay report
N1b	Release / replay timeline
N2	Gishboliner–Li: Pach–Tardos
N3	Wu: operator cotype counterexample
N4	Keevash: Kahn flow conjecture
N5	Su–Hu: negative dependence
N6	Braun: bit-PHP in Res(xor)
N7	FormalFlow: long-horizon autoformalization
N8	Gardam lattice: unique products

News scope retained from the supplied deck: September 17–23, 2026. See the source links for current status.

Source index 02 | Readings and Lean

N9	Liu: magnitude continuity
N9a	AI-disclosure tracker (secondary)
N10	Grad counterexamples
N11	Primus for Math: product announcement
R1	Gowers's Weblog: reading 01
R1z	Zhihu: supplied article reference
R2	OpenAI: reading 02
R3	Independent advisory group
L1	Theorem Proving in Lean 4: Inductive Types
L2	Lean 4.19 reference: Quotients
L3	Theorem Proving in Lean 4: Axioms and Computation

The two original articles open from slides 7 and 10; full article text is not reproduced in this deck.